



Africa Nighttime Lights Atlas

2012 - 2024





Data Innovation for Africa



EARTHDATA



AFRICA
NIGHTTIME LIGHTS ATLAS

Illuminating
Africa's Future:
Turning
Spaceborne
Light Into
Development
Power



NASA's nighttime lights



NASA's nighttime lights (NTL) imagery has a wide range of uses, including tracking urban growth, assessing electrification, monitoring disasters, and studying the biological impacts of light pollution.

<https://ladsweb.modaps.eosdis.nasa.gov/missions-and-measurements/science-domain/nighttime-lights/>



VIIRS Black Marble



● Name & acronym

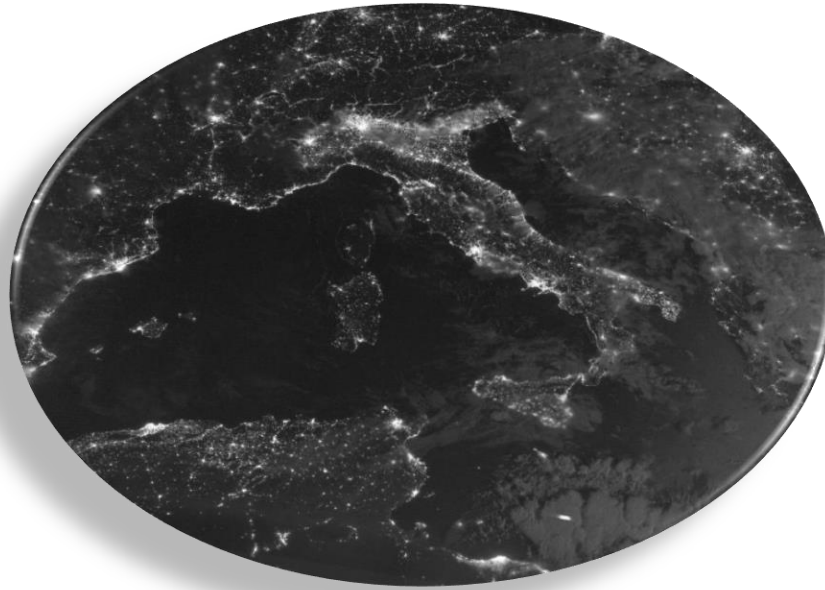
- *VIIRS = Visible Infrared Imaging Radiometer Suite*
- *Black Marble = NASA's global product for nighttime lights (NTL)*

● Purpose

- Measures *nighttime radiance* on Earth to monitor human activity, electrification, urbanization, and environmental change.

● Platform

- Onboard the *Suomi NPP* and *NOAA-20* satellites
- Operated under NASA and NOAA



<https://blackmarble.gsfc.nasa.gov/>

● Unit of measurement

- *Radiance* in $\text{nanowatts} / \text{cm}^2 / \text{sr}$ ($\text{nW} \cdot \text{cm}^{-2} \cdot \text{sr}^{-1}$)
→ Standard physical unit for measuring emitted light intensity

● Spatial resolution

- *~500 meters* at nadir

● Temporal resolution

- *Daily* observations
- *Monthly and annual* composites available

● Key corrections applied

- Cloud masking
- Lunar illumination correction
- Atmospheric correction
- Terrain correction
- Stray light removal
- Snow masking

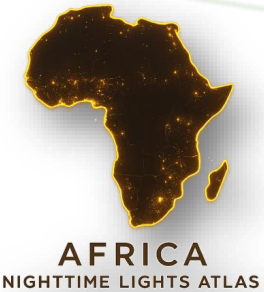
Who conducted this work?

Data Innovation for Africa – AfDB

- Led by **Data Innovation for Africa (DI4A)**.
- Driven by the **Data Innovation Lab (DIL)**.
- Uses **NASA Black Marble** and other advanced datasets.
- Fully aligned with **AfDB's Digital Ambition 2030**.
- A **continental program** covering all **54 African countries**.



Nighttime Lights Data – Datasets sources



The analysis relies on NASA’s VIIRS Black Marble products—globally consistent daily, monthly, and annual nighttime light composites available since 2012.

Product Name	Suomi NPP
Daily Gridded Day Night Band 500m Linear Lat Lon Grid Night	VNP46A1 ▾
Gap-Filled Lunar BRDF-Adjusted <u>Nighttime Lights Daily</u> L3 Global 500m Linear Lat Lon Grid	VNP46A2 ▾
Lunar BRDF-Adjusted <u>Nighttime Lights Monthly</u> L3 Global 15 arc second Linear Lat Lon Grid	VNP46A3 ▾
Lunar BRDF-Adjusted <u>Nighttime Lights Yearly</u> L3 Global 15 arc second Linear Lat Lon Grid	VNP46A4 ▾

Insights generated for all 54 african countries (1/5)



Country-level reports (54 Countries)

Complete NTL profiles covering 13 years (2012-2024), with national, regional, and local-level trends.

Multi-level administrative analysis

Insights for Admin-0, Admin-1, and Admin-2/3, showing disparities and development hotspots.

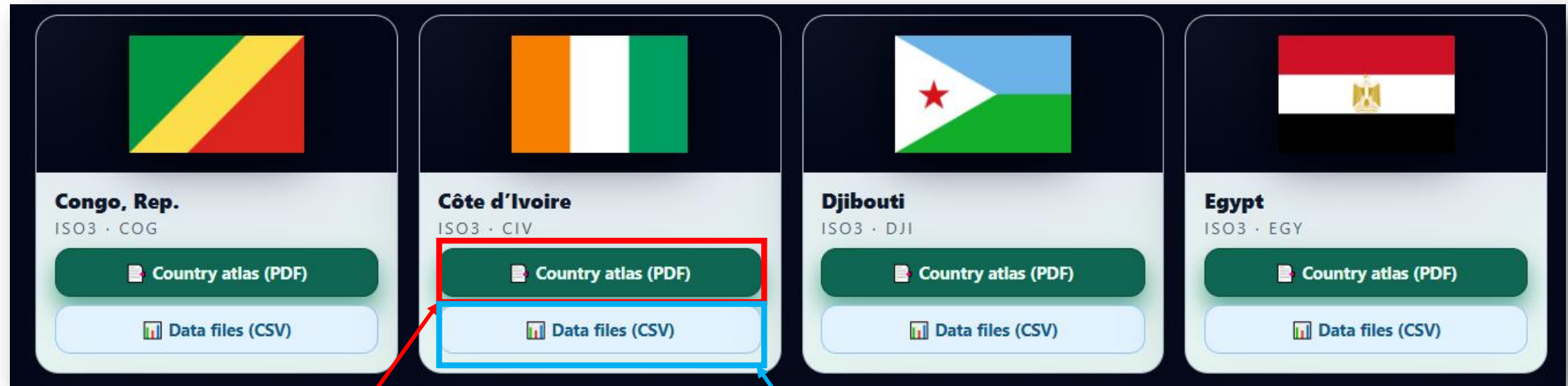
Standardized & comparable indicators

Harmonized radiance metrics enabling comparisons across countries, regions, and time.

Ready-to-use country datasets

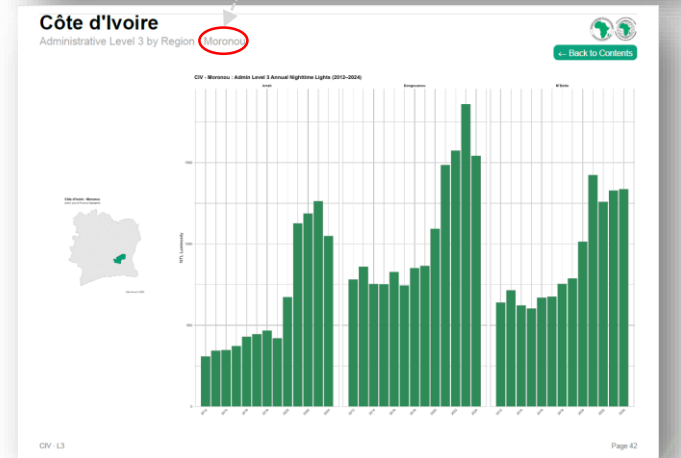
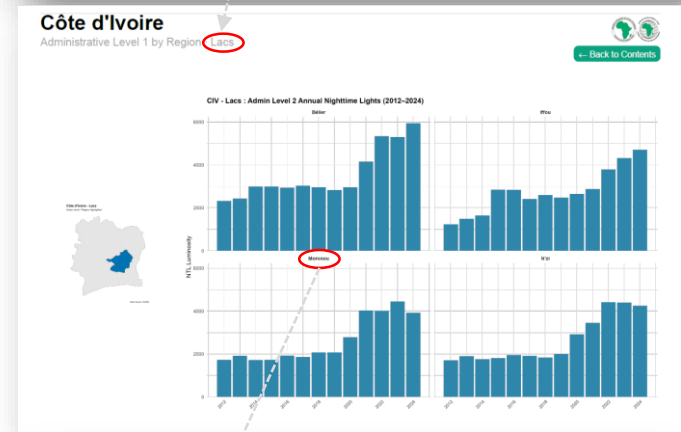
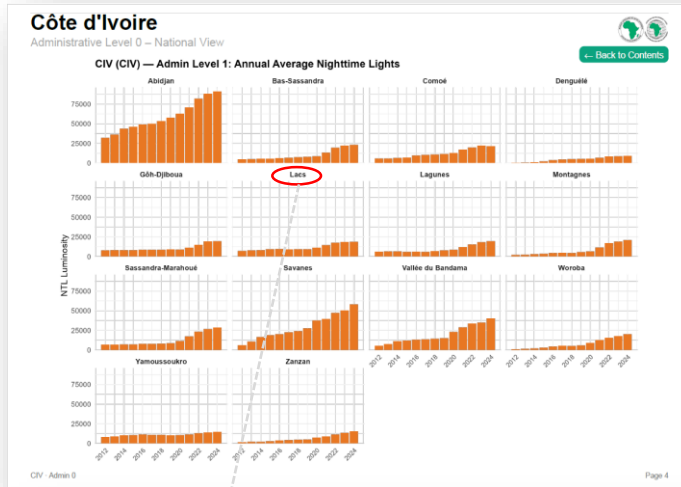
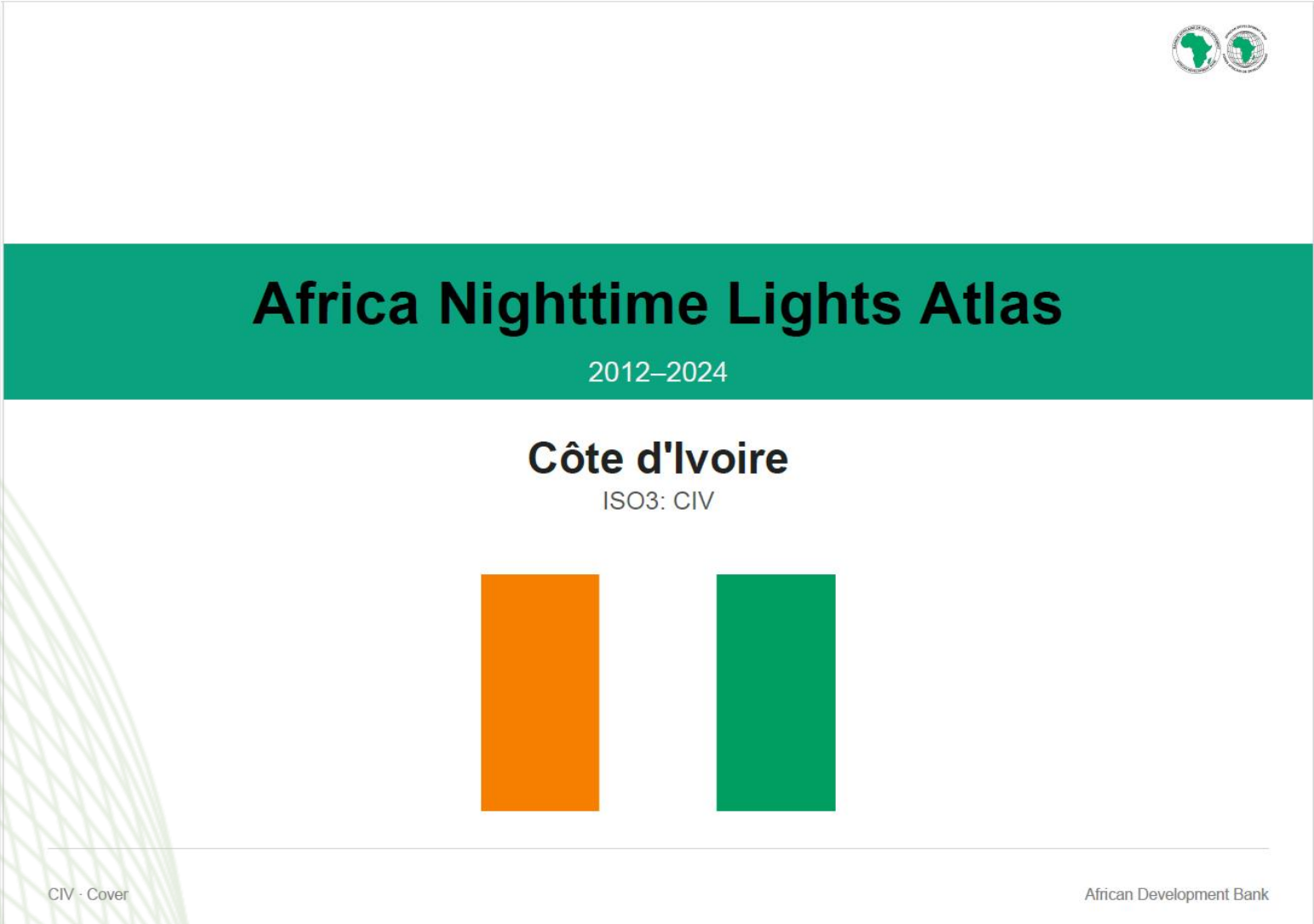
A complete dataset for all 54 African countries, organized by administrative levels (Admin1–Admin3) and delivered in clean, ready-to-use CSV files covering 2012–2024.

Insights Generated for All 54 African Countries (2/5)



Access to PDF Report

Access to Data



Insights Generated for All 54 African Countries (4/5)

Côte d'Ivoire – Nighttime Lights Data

Files

Annual VIIRS Black Marble statistics for Côte d'Ivoire.

COUNTRY CODE · CIVFORMAT · CSVADMIN LEVELS 1–3

← Back to main atlas

Available CSV files

Annual nighttime lights intensity aggregated at different administrative levels.

Admin 1

CSV · 2012–2024
CIV_Admin1_NTL_Annual_2012_2024.csv
First administrative level

↓ Download

Admin 2

CSV · 2012–2024
CIV_Admin2_NTL_Annual_2012_2024.csv
Second administrative level

↓ Download

Admin 3

CSV · 2012–2024
CIV_Admin3_NTL_Annual_2012_2024.csv
Third administrative level

↓ Download

Generated by the **Data Innovation Lab (DIL)** of the **African Development Bank (AfDB)** – Africa Nighttime Lights Analytics Program.
Contact: A.AIH@afdb.org

Download Data



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DE DEVELOPEMENT

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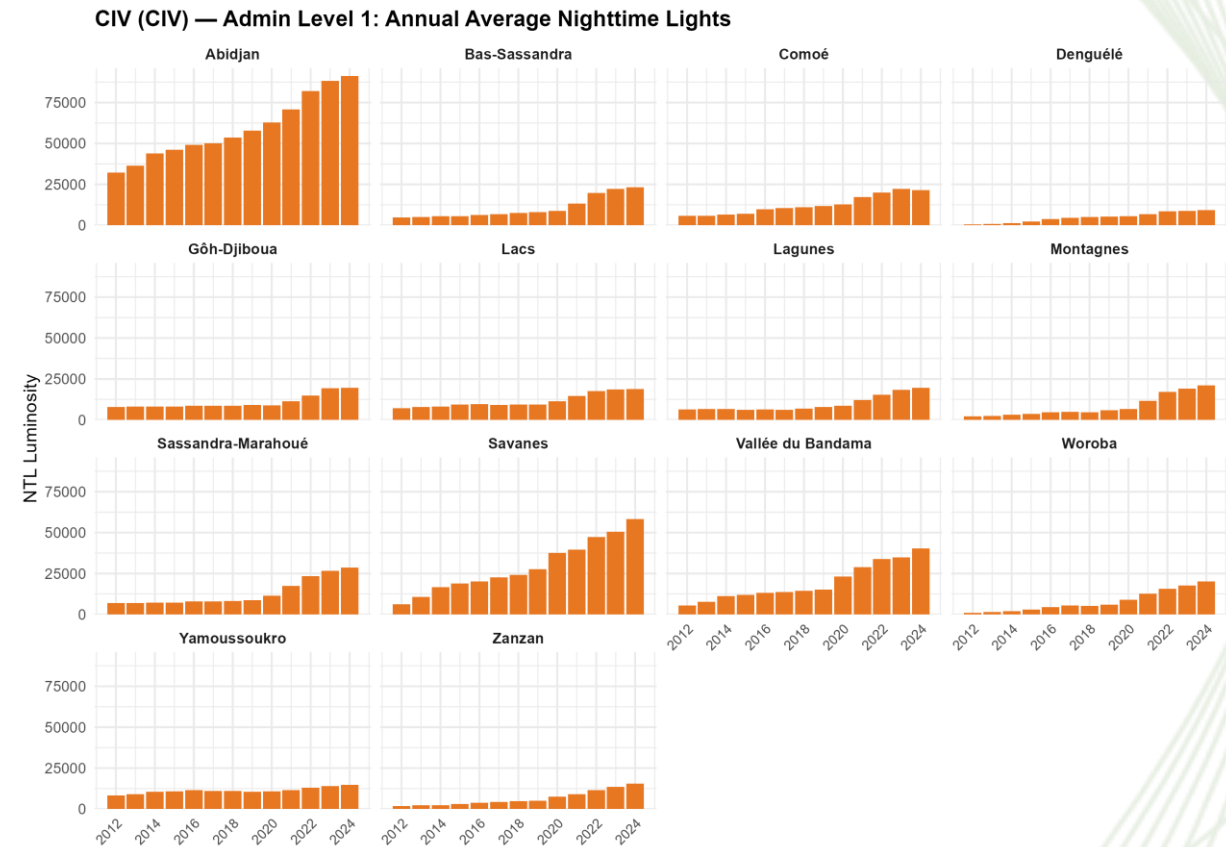
Insights Generated for All 54 African Countries (5/5)

	A	B	C	D	E
1	NAME_1	NAME_2	NAME_3	date	ntl_mean
2	Abidjan	Abidjan	Abidjan	2012	32263.935546875
3	Abidjan	Abidjan	Abidjan	2013	36344.96875
4	Abidjan	Abidjan	Abidjan	2014	43965.171875
5	Abidjan	Abidjan	Abidjan	2015	46011.49609375
6	Abidjan	Abidjan	Abidjan	2016	49141.25390625
7	Abidjan	Abidjan	Abidjan	2017	50075.6640625
8	Abidjan	Abidjan	Abidjan	2018	53628.6328125
9	Abidjan	Abidjan	Abidjan	2019	57950.12109375
10	Abidjan	Abidjan	Abidjan	2020	62938.390625
11	Abidjan	Abidjan	Abidjan	2021	70679.15625
12	Abidjan	Abidjan	Abidjan	2022	82080.7734375
13	Abidjan	Abidjan	Abidjan	2023	88206.078125
14	Abidjan	Abidjan	Abidjan	2024	91337.109375
15	Bas-Sassandra	Gbôkle	Fresco	2012	114.186721801758
16	Bas-Sassandra	Gbôkle	Fresco	2013	94.0883331298828
17	Bas-Sassandra	Gbôkle	Fresco	2014	118.746520996094
18	Bas-Sassandra	Gbôkle	Fresco	2015	117.162467956543
19	Bas-Sassandra	Gbôkle	Fresco	2016	113.998924255371
20	Bas-Sassandra	Gbôkle	Fresco	2017	110.895317077637
21	Bas-Sassandra	Gbôkle	Fresco	2018	118.440322875977
22	Bas-Sassandra	Gbôkle	Fresco	2019	117.108955383301
23	Bas-Sassandra	Gbôkle	Fresco	2020	137.26220703125
24	Bas-Sassandra	Gbôkle	Fresco	2021	366.776580810547
25	Bas-Sassandra	Gbôkle	Fresco	2022	738.4814453125
26	Bas-Sassandra	Gbôkle	Fresco	2023	948.039306640625
27	Bas-Sassandra	Gbôkle	Fresco	2024	1009.39874267578
28	Bas-Sassandra	Gbôkle	Sassandra	2012	233.038879394531
29	Bas-Sassandra	Gbôkle	Sassandra	2013	278.24472045898

CIV_Admin3_NTL_Annual_2012_2024.csv

Understanding the scale of current NTL values

- The values represent the **total annual nighttime radiance** for each administrative region.
- They are calculated by **summing the radiance of all VIIRS pixels** within each region.
- Large numbers (10,000–70,000+) are normal because regions contain **thousands of pixels**, each contributing light intensity.
- The annual VNP46A4 product already aggregates multiple nightly observations, further increasing totals.
- These values allow a **realistic comparison of lighting intensity and economic activity** across regions within each country and over time.



Policy & Development Domains Informed by NTL Insights

Population distribution and density

Identify inhabited areas and their density

Impact of conflicts, crises, and displacement

Detect deserted or displaced areas

Urbanization and city growth

Monitor the expansion of urban areas

Economic activity and development

Estimate the intensity of economic activity

Access to energy and electrification

Measure access to electricity



Predictable demographic growth

Anticipate population increases

Infrastructure and transport

Assess the effects of government projects

Impact of public policies

Evaluate the effects of government projects

Post-Crisis Economic Recovery

Observe the recovery after a crisis

Environmental Monitoring and Energy Waste

Identify energy losses and excessive lighting

Reveal the Power of Data in Africa



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